

# Spontaneous Dialect-Aware Speech Corpus for Low-Resource Dakhini, A Southern Indo-Aryan Language: Methods, Challenges, and Insights

Anindita Mondal <sup>1,\*</sup>, Priyanka Kommagouni<sup>2,\*</sup>

<sup>1</sup> Language Technologies Research Center, IIIT Hyderabad, India

<sup>2</sup> Independent Researcher

anindita.mondal@research.iiit.ac.in, parvathipriyanka86@gmail.com

## Abstract

Dakhini, a contact variety spoken in the Deccan region of India, remains severely underrepresented in curated speech resources despite widespread conversational use. This work presents a data-centric study aimed at informing systematic Dakhini speech corpus development. Using telephonic conversational recordings, we examine factors relevant to corpus development, including context-driven code-switching and speaker variation across gender and educational backgrounds. Phonological reductions, auxiliary omission, and participial variation are observed in spontaneous speech, while formal contexts trigger shifts toward standardized Hindi/Urdu forms. These patterns highlight the importance of speaker stratification and dialect-aware annotation when curating low-resource conversational datasets. Our findings provide guidelines for developing Dakhini speech corpora and lay groundwork for dialect-sensitive ASR and conversational modeling in under documented language varieties.

**Index Terms:** Dakhini, low-resource, spontaneous speech, dialect-aware

## 1. Introduction

Speech data collection for well-resourced languages has benefited from decades of established methodology, standardised protocols, and large speaker pools. However, when the target variety is a rare or endangered dialect - particularly one that exists primarily as a spoken, community-bound phenomenon with no formal written standard - the data curation pipeline faces a qualitatively different class of challenges. This paper examines these challenges through the lens of Dakhini (Hyderabadi Hindi), a dialect spoken in the Deccan region of southern India that blends Persian, Old Urdu (Dehlavi), Kannada, Marathi, and Telugu. We argue that robust data collection for such varieties demands thorough structural and sociolinguistic understanding of the dialect before any recording session begins.

Dakhini presents a compelling case study precisely because of its complex sociolinguistic standing. It is not codified in formal educational systems, has no widely accepted written form in modern use, and its speakers exist along a continuum ranging from full Dakhini-dominant speech to varying degrees of code-switching with standard Hindi/Urdu. A prior survey of 60 speakers demonstrated that speaker behaviour shifts measurably between informal and formal settings, with women in particular showing a stronger tendency to shift toward standard Hindi/Urdu with Dakhini features, and educated speakers of both genders doing the same [1]. This is not an incidental observation: it is a fundamental data collection risk. If a speaker

perceives a recording session as formal or evaluative, they may unconsciously suppress the very dialect features the researcher intends to capture. This phenomenon, well-documented in sociolinguistics as the Observer’s Effect [2], is significantly amplified in the case of non-prestige dialects. Speakers of Dakhini, like speakers of many regional dialects without official recognition, have internalised the association between their dialect and informality, intimacy, or lower social register. Faced with a microphone, a researcher, or an institutional framing, they may perform a more standard variety rather than producing naturalistic dialect speech [3]. The result is a dataset that is acoustically clean but linguistically inauthentic, systematically missing the phonological and morphosyntactic features that define the dialect.

What distinguishes this problem from general data quality concerns is that the drift cannot be identified or corrected without prior structural knowledge. For a collector unfamiliar with Dakhini phonology, the reduction of long vowels (e.g., /aadmi:/ → /admi:/), consonant assimilation patterns (e.g., /itnaa/ → /ittaa/), the characteristic non-use of the auxiliary *hai*, or the substitution of *mere se* for *mujh se* might go unnoticed or might even be treated as speaker error. In reality, these are diagnostic features. Their presence or absence is precisely what distinguishes genuine Dakhini from a Dakhini-inflected standard Hindi. A structural and phonological checklist, derived from careful prior linguistic analysis, is therefore not merely helpful for data collection, it is the primary quality control mechanism.

The preliminary structural investigation covered Dakhini’s phonology (consonant assimilation, vowel-length reduction), morphology (noun and adjective classes), pronoun system, and verbal constructions, documenting systematic divergences from standard Urdu. These findings directly informed participant selection, elicitation design, and the annotation rule sets described in this paper. This paper makes three contributions: (i) articulating the specific sociolinguistic and structural challenges of collecting speech data from a non-prestige dialect community; (ii) proposing a principled dialect-aware data collection and annotation methodology grounded in prior structural analysis; and (iii) describing elicitation and recording design strategies that maximise authentic dialect production rather than monitored speech.

## 2. Sociolinguistic Background and the Role of Preliminary Analysis

Dakhini and Urdu share a substantial lexical and phonological base, therefore, distinguishing between the two in spontaneous speech is not always straightforward. However, certain identifying features consistently mark Dakhini usage: characteristic lexical items such as *kaiku*, *nakko*, and *manjhe*; phonological

\*These authors contributed equally.

tendencies like consonant assimilation and vowel shortening; and grammatical constructions such as the reduced use of the auxiliary *hai* or the substitution of *ko* in place of *kar* in sequential actions [4]. The preliminary study systematically documented these features to develop a practical checklist that could assist researchers in identifying authentic Dakhini speech during telephonic recordings [5]. Rather than relying solely on speaker self-identification, linguistic markers became crucial tools for validation. The preliminary survey also highlighted the role of socioeconomic background and gender in shaping dialect use. Speakers from economically disadvantaged backgrounds demonstrated greater comfort using Dakhini even in relatively formal contexts, whereas individuals from educated sectors more frequently shifted toward standardised language [6]. Out of 15 male participants from economically disadvantaged backgrounds, 10 were comfortable using Dakhini in formal settings, while 11 female participants from the same group also used Dakhini in formal contexts. By contrast, among educated-sector participants, only 6 out of 15 male participants and just 2 female participants used Dakhini in formal settings, with the majority defaulting to standard Hindi/Urdu. These insights informed participant selection strategies, ensuring representation across social categories while anticipating likely patterns of code-switching.

### 3. Motivation

Despite its historical depth and sociolinguistic richness, spontaneous spoken Dakhini remains underexplored, with most research focusing on its historical evolution, literary tradition, or structural description in controlled contexts [5]. These studies affirm Dakhini’s distinctiveness but overlook its function in natural, contemporary speech, especially in mediated settings like telephonic communication. The dialect’s structural closeness to standard Urdu often leads to its marginalisation as a regional variant, complicating the identification of authentic usage and risking the inclusion of code-switched or standardised forms in spontaneous speech data. Moreover, earlier work rarely integrates sociolinguistic variables such as gender, education, and context sensitivity into methodological design. Addressing these gaps, the present study develops an empirically grounded, context-sensitive checklist for participant selection, conversational prompting, and real-time dialect identification, ensuring corpus authenticity while contributing to the preservation and documentation of Dakhini as a living, evolving linguistic heritage.

### 4. Conversational Data Collection: Design, Ethics, and Speaker Recruitment

Conversational speech data were collected through telephonic interactions as illustrated in Figure 1 between pairs of participants who were already familiar with each other, ensuring ecological validity and minimising the social distance that typically triggers code-switching. Each pair was composed of one male and one female speaker. This design served a dual purpose: it reflects naturalistic interactional contexts in Dakhini-speaking communities, and the acoustic contrast between male and female voices significantly reduces speaker confusion during automated diarization, improving segmentation reliability on spontaneous conversational data.

To further reduce the Observer’s Effect, an asymmetric awareness protocol was adopted. Within each pair, only one participant, referred to as the anchor speaker, was informed

of the recording setup prior to the conversation. The anchor speaker was briefed to lead the conversation naturally, maintain turn-taking, and steer discussion toward a pre-agreed topic menu when needed, without revealing the recording context to their interlocutor. This approach follows the established sociolinguistic practice of covert recording with retrospective consent, a methodology grounded in Labov’s foundational work [2], which demonstrated that prior awareness of recording systematically alters speaker behaviour in ways that undermine dialectal authenticity.

The corpus comprises recordings from 160 speakers from Hyderabad, India, including approximately 10% from early-settler families. Participants range in age from 17 to 50 years, with a gender distribution of 70% male and 30% female. The cohort spans diverse educational and socioeconomic backgrounds, with approximately 20% having higher educational attainment and 20% belonging to upper socioeconomic strata, while the majority represent middle- and lower-income groups. Speakers are multilingual, with varying proficiency in Dakhini, Urdu, Telugu, and English. Each recording session lasted approximately 20 minutes, from which about 5 minutes of representative conversational speech were selected for inclusion in the corpus.



Figure 1: *Recruitment workflow for collecting conversational Dakhini speech, including speaker identification, community briefing, pair formation, and anchor briefing.*

Following each conversation, the uninformed participant was fully debriefed and the research intent was explained clearly, that the goal was to build a linguistically rigorous corpus of Dakhini to support its recognition in academic research and language technology, and that their contribution as a native speaker was foundational to this work. Informed consent was then sought explicitly before any data was retained. Recordings were accepted into the corpus only upon explicit consent; sessions where consent was withheld were discarded in their entirety with no data retained. Participants were also informed of their right to withdraw their data at any stage. Post-consent, diarization was applied to retain only the audio of the target speaker, as these segments represent the most unmonitored dialect production. The anchor speaker’s audio was excluded from the primary corpus.

Topic control was introduced to address a practical challenge encountered in pilot recordings: fully unstructured conversation, while naturalistic, produced sessions dense with interjections, tangential corrections, and topic drift that complicated processing and reduced usable dialectal content. Rather than scripting interactions, anchor speakers were provided a loose menu of culturally resonant, casual topics such as a recently released film, a new restaurant opening, celebrity news, or a recipe exchange such as preparing biryani. This guided but informal design preserved spontaneity while keeping sessions tractable. Speaker recruitment required careful and transparent communication. Dakhini speakers, particularly those with formal education, often carry an internalised stigma around dialect

use, associating it with informality or lower prestige.

## 5. Challenges in Data Collection and Community Engagement

One of the earliest challenges in this project was the identification of authentic Dakhini speakers, who are not uniformly distributed across Southern India, but concentrated in specific pockets of the older neighbourhoods, where intergenerational dialect transmission has remained relatively intact. Conventional open recruitment strategies systematically missed this population, attracting speakers whose speech showed substantial convergence toward standard varieties, making hyperlocal community-embedded outreach through trusted neighbourhood intermediaries essential [6]. Effective recruitment ultimately required working through existing social networks and local contacts within specific neighbourhoods to identify speakers whose dialect use was consistent and naturalistic across both formal and informal contexts [7].

Even once identified, recruiting speakers was far from straightforward. The contemporary data market has generated well-founded wariness in many communities around how recordings would be used, stored, and accessed. Scepticism layered onto the internalised stigma that many Dakhini speakers already carry toward their own dialect [1]. This made individual solicitation largely ineffective and, more problematically, tended to produce participants who agreed to record but then monitored their speech heavily, defeating the purpose of the collection entirely. Rather than individual solicitation, the recruitment strategy was restructured around community information sessions, where small groups of prospective participants engaged in open dialogue with the research team about the project’s linguistic objectives and methodology. These sessions reframed participation as collective contribution, an opportunity for speakers to play an active role in bringing their dialect into formal linguistic scholarship and language technology, with the message that without their voices, Dakhini would remain invisible to the tools and systems that increasingly mediate daily life.

## 6. Database Preparation Methodology

### 6.1. Recording Modality: Prioritising Linguistic Authenticity Over Acoustic Control

Telephonic interaction was chosen over studio recording as a deliberate methodological decision. Studio environments introduce the formal, evaluative context that most reliably triggers dialect suppression in speakers of non-prestige varieties. The telephone is a channel already associated in speakers’ daily lives with casual, intimate exchange - and for the uninformed participant, the interaction was indistinguishable from an ordinary call, eliminating any behavioural cue that could prompt self-monitoring. The resulting trade-off in acoustic quality was conscious and considered: reduced signal fidelity was accepted in exchange for linguistic authenticity, which for dialect documentation is the more consequential variable.

Unlike face-to-face interaction, telephonic exchanges lack visual cues and often encourage more spontaneous, everyday language use. This property was intentionally leveraged. The preliminary study thus also functioned as a methodological rehearsal, helping define appropriate conversation themes, identify dialect-sensitive lexical triggers, anticipate shifts toward Urdu, and establish criteria for evaluating authenticity in real-time data.

### 6.2. Diarization

The conversational recordings were processed through a structured speech diarization (pyannote speaker-diarization-3.1 [8]) pipeline consisting of three principal stages: (i) extraction of speaker embeddings to capture speaker-specific acoustic characteristics (ECAPA-TDNN embeddings [9] trained on VoxCeleb [10] dataset), and (ii) clustering-based segmentation to assign speaker labels to contiguous speech segments. The diarized output, in the form of Rich Transcription Time Marked (RTTM) files, was subsequently reviewed for segmentation accuracy. During the chunking stage, a gradual amplitude ramp-up and ramp-down was applied at the boundaries of each segment so that abrupt cuts do not introduce audible artefacts in the concatenated speech. Segments corresponding to each speaker were then concatenated to generate individual speaker-level audio files. This dual representation - retaining the original conversational structure while producing speaker-specific compilations - facilitates both interactional analysis and speaker-level linguistic investigation.

Given the spontaneous and dialect-rich nature of the data, particular attention was paid to diarization robustness. Natural conversations often include overlapping speech, backchannels, and rapid turn exchanges, which challenge automated segmentation systems. A manual verification stage was therefore incorporated to correct speaker misattributions and boundary inconsistencies. Corrected outputs were archived to support iterative refinement of the diarization models, thereby improving system performance on dialectal conversational speech over time. The mixed-gender pair design also contributed to improved diarization accuracy.

### 6.3. Annotation Framework: Semi-Automated Workflows with Dialect-Aware Rule Sets

A central objective of the project is to streamline the annotation process through semi-automated workflows while maintaining high levels of linguistic reliability. Since most available ASR systems are trained on standard Hindi or Urdu, their outputs systematically misrecognise or normalise Dakhini-specific forms, transcribing, for instance, the Dakhini /ittaa/ as the standard /itnaa/, or missing dropped auxiliaries entirely. In our work, we used IndicConformer Hindi ASR <sup>1</sup>. Rather than treating these outputs as errors, the annotation pipeline, as shown in Figure 2, was designed to treat ASR-to-standard mismatches as positive dialect signals: divergences between the ASR transcript and the audio are flagged as candidate sites for dialect-specific annotation rather than simple transcription errors. Drawing directly from the structural analysis of Dakhini, the following rule-based detection modules were implemented as listed in Table 1. The following abbreviations were adopted for the tags:

- **PA**: Phonological Assimilation
- **PV**: Phonological Vowel Length Reduction
- **MAUX**: Morphosyntactic Auxiliary Omission
- **MP-KO**: Morphosyntactic Participle “ko” Construction
- **MP**: Morphosyntactic Pronoun Variation
- **SF**: Sociolinguistic Formality Register Marker
- **LD**: Lexical Dakhini-specific Item

The Consonant Assimilation Detector (PA) flags segments where /n/ preceding or following /t/ is assimilated, surfacing forms such as /ittaa/ for /itnaa/ and voicing assimilation pat-

<sup>1</sup><https://github.com/AI4Bharat/IndicConformerASR>

Table 1: Mapping of Standard Hindi/Urdu Forms to Dakhini Variants with Corresponding Rule-Based Detection Tags.

| Standard Hindi/Urdu Form                | Dakhini Realization | Tag   |
|-----------------------------------------|---------------------|-------|
| <b>Phonological Rules</b>               |                     |       |
| itnaa (how much)                        | ittaa               | PA    |
| jitnaa (however much)                   | jittaa              | PA    |
| itnaa hi (that’s all)                   | ittaich             | PA    |
| us zamaane (in those days)              | uzzamaane           | PA    |
| aadmii                                  | admii               | PV    |
| acchaa                                  | accha               | PV    |
| bataao                                  | batau               | PV    |
| DhuunD                                  | Dhund               | PV    |
| <b>Morphosyntactic Rules</b>            |                     |       |
| (wo jaa rahe hai)                       | woh jaa rahe        | MAUX  |
| usne diye the                           | une diye            | MAUX  |
| woh gaye hai                            | woh gaye            | MAUX  |
| main khaa kar aaya                      | main khaa ko aaya   | MP-KO |
| mujh se                                 | mere se             | MP    |
| unheN                                   | une                 | MP    |
| inheN                                   | ine                 | MP    |
| aap                                     | tum / tu            | SF    |
| <b>Code-Switching and Lexical Layer</b> |                     |       |
| kyon (why)                              | kaiku               | LD    |
| mat karo (don’t)                        | nakko karo          | LD    |
| mein / ke andar (in / among)            | manjhe              | LD    |
| dhire se (slowly / gently)              | haule               | LD    |

terns such as /uzzamaane/ for /us zamaane/. The Vowel Length Reduction Detector (PV) flags shortened vowels in unaccented syllables : /admii/ for /aadmii/, /accha/ for /acchaa/, /batau/ for /bataao/. The Auxiliary Omission Detector (MAUX) flags verb forms appearing without an expected auxiliary, e.g., /une diye/ (“he gave”), /woh gaye/ (“they left”), one of the most diagnostically significant features of Dakhini, requiring parse-level rather than string-level matching. The Participle Construction Detector (MP-KO) flags the Dakhini-specific use of ko in place of standard kar in sequential action constructions. The Pronoun Form Detector (MP) identifies *mere se* for *mujh se* and neutralisation of *ine/une* distinctions through optional /h/ and /N/ deletion in rapid speech. The Formality Register Detector (SF) flags *tum* or *tu* in contexts where standard Hindi/Urdu would require *aap*, a feature that disappears in speakers with formal Hindi schooling and therefore serves double duty as both a structural and a sociolinguistic register marker. Finally, a dedicated Lexical Tagger (LD) was built around a seed vocabulary of Dakhini-specific imports from regional languages such as *kaiku* (“why”), *nakko* (“don’t”), *manjhe* (“in/among”), *haule* (“slowly”). These items that have no equivalent form in standard Urdu and whose presence serves as a macro-level authenticity indicator across the corpus.

Rather than applying uniform human review across all segments, the pipeline uses tag density as a triage signal. Segments receiving multiple overlapping tags, simultaneously flagged for, say, PA, MAUX, LD: are prioritised for detailed human review, as they represent the highest concentration of dialectal activity and carry the greatest risk of compounded ASR error. Conversely, segments with no dialect flags are passed through with lighter review, allocating annotator effort where it is most consequential. Annotators were trained using standardised guide-

lines derived directly from the structural analysis, with worked examples for each tag category. Inter-annotator agreement [11] was measured on a held-out validation set, with disagreements used iteratively to refine rule boundaries and update annotator guidelines.

To ensure methodological rigour, systematic quality control procedures were incorporated into the annotation pipeline. Inter-annotator agreement metrics were employed to measure consistency, particularly for dialect classification and code-switching identification. Periodic blind re-annotation of randomly selected samples was conducted to detect annotation drift and maintain standardisation over time. Automated outputs were benchmarked against manually curated reference segments to quantify transcription and tagging accuracy. The modular design of the pipeline enables iterative improvement: corrections introduced during human validation are fed back into the automated models, gradually enhancing performance through supervised refinement. Metadata tagging, including participant background, conversational topic, and recording conditions were systematically maintained to support sociolinguistic analysis and facilitate reproducibility.

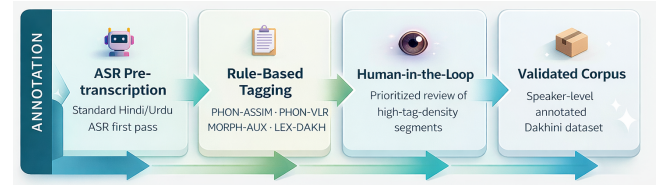


Figure 2: Annotation workflow for the Dakhini corpus involving ASR-based pre-transcription, rule-based tagging, human verification, and final corpus creation.

## 7. Conclusion

This paper presented a data collection and curation methodology for Dakhini, a historically rich but critically under-resourced dialect of the Deccan region. We demonstrated that authentic dialect documentation demands far more than recording infrastructure - it requires deep structural knowledge of the variety, culturally sensitive community engagement, and deliberate design choices across the entire pipeline, from elicitation and recording modality to diarization and annotation. The dialect-aware rule sets and asymmetric awareness protocol introduced here offer replicable strategies for researchers working with similarly situated varieties. We hope this work contributes both a practical framework and a broader argument: that linguistic authenticity must be treated as a first-order design constraint in dialect corpus construction, not an afterthought, and that the collector’s linguistic competence is as important as the recording equipment. We provide a few audio samples here <sup>2</sup>.

## 8. Generative AI Use Disclosure

Generative AI tools were used to assist with language refinement and diagram formatting during the preparation of this manuscript. All intellectual contributions, experimental design, analysis, and final content decisions were made by the authors, who take complete responsibility of the work.

<sup>2</sup><https://mondal-anindita.github.io/Dakhini/>

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